

Memo: Switching to Recycled Cotton

To: Sourcing & Sustainability Leads **From:** @Someone **Date:** 2026-09-18

Recommendation: switch to recycled cotton

Switching from virgin (conventional) to recycled cotton fiber cuts the cradle-to-gate carbon footprint by roughly **87%** and water consumption by **78-100%** per kilogram, across two independent life-cycle assessments. A peer-reviewed academic LCA on cotton yarn corroborates the direction and the water figure closely (within 25% on liters saved per kilogram), even though its absolute numbers differ because it measures a wider production boundary.

In practice, 100% recycled cotton fiber is rarely used -- mechanical recycling shortens the fiber, so commercial yarns are usually blended with virgin cotton. At a realistic **80% recycled-content blend**, documented savings are still strong: **~78% less water** and **~57% less energy** than an all-virgin equivalent, with GHG savings estimated at roughly **70%** on the same basis.

The numbers

Metric	Source	Virgin cotton	Recycled cotton	Reduction
GHG (kg CO2e/kg fiber)	Recover(tm) verified LCA, 2022	1.99	0.26	87%
Water (L/kg fiber)	Recover(tm) verified LCA, 2022	--	--	2,116 L saved/kg
GHG (kg CO2e/kg yarn)	Liu et al. 2020, Int J LCA	--	--	6.6 saved/kg
Irrigation water (L/kg yarn)	Liu et al. 2020, Int J LCA	--	--	2,783 L saved/kg
Energy, 80% recycled blend	Ecotec(r) LCA (Marchi & Fildi)	--	--	56.6%

Metric	Source	Virgin cotton	Recycled cotton	Reduction
Water, 80% recycled blend	Ecotec(r) LCA (Marchi & Fildi)	--	--	77.9%

The fiber-level figure (Recover) and the yarn-level figure (Liu et al.) measure different production stages, so their absolute numbers should not be averaged -- but they agree closely on water savings per kilogram, which is a genuine cross-check pass.

Worked example: a 10,000 kg fiber order placed at an 80% recycled-content blend saves an estimated **13.8 tCO₂e** and **16,928 m³ of water** versus the same order placed 100% virgin -- roughly the annual water use of 100+ households. Full calculation in the companion Excel model's "Blended Adoption & Calculator" tab, where the order quantity and blend % can be changed.

Why a blend, not 100% recycled

Mechanical recycling breaks and shortens cotton fibers, which weakens the yarn. Most commercial recycled-cotton yarns therefore blend 20-80% recycled fiber with virgin cotton to stay spinnable at production scale. The 87% GHG / near-100% water figures above are the theoretical ceiling for pure recycled fiber; the 57-78% figures at an 80% blend are what can actually be ordered today. Either way, the direction and the scale of the saving hold.

Bottom line and next step

Switching a cotton order from virgin to recycled content is one of the highest-leverage single material substitutions available to a sourcing team: an 80% recycled blend realistically cuts water use by ~78% and energy by ~57% per kilogram, with GHG savings in the same range, at commercially available blend ratios today.

Recommended next step: pilot an 80% recycled-content blend on one product line's next fiber order, using the Excel calculator to size the water and carbon savings for that specific order quantity, and confirm yarn strength/quality meets spec before scaling further.

Sources

- [Recover\(tm\): What is Recycled Cotton and Why Does it Matter?](#) -- verified LCA, EcoReview 2022, citing Higg MSI v3.6
- Liu, Y. et al. (2020). "Could the recycled yarns substitute for the virgin cotton yarns: a comparative LCA." *The International Journal of Life Cycle Assessment*, 25(10), 2050-2062
- Ecotec(r) recycled cotton yarn LCA, Marchi & Fildi (cradle-to-gate, commercial blend ratios)